

Pattern 21 :- $\times a \pm b$ (b change in pattern)

Rule :- Multiply by a fixed number ; add/subtract values that follow a pattern.

Examples,

$$\begin{array}{ccccccccc}
 1 & , & 3 & , & 9 & , & 23 & , & 53 & , & 115 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 \\
 + 1 & & + 3 & & + 5 & & + 7 & & + 9 & &
 \end{array}$$

$$\begin{array}{ccccccccc}
 10 & , & 28 & , & 80 & , & 234 & , & 694 & , & 2072 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 \\
 - 2 & & - 4 & & - 6 & & - 8 & & - 10 & &
 \end{array}$$

Practice Questions:

Q1. 6, 20, 64, 198, 602, ?

(A) 1816

(B) 1856

(C) 1846

(D) 1796

(E) 1811

$$\begin{array}{ccccccccc}
 6 & , & 20 & , & 64 & , & 198 & , & 602 & , & (?) \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 & & \times 3 \\
 + 2 & & + 4 & & + 6 & & + 8 & & + 10 & &
 \end{array}$$

$(602 \times 3) + 10$
 $= 1806 + 10$
 $= 1816$

Q2. ?, 15, 32, 67, 138, 281

(A) 9

(B) 12

(C) 10

(D) 11

(E) 7

$$\begin{array}{ccccccccc}
 (?) & , & 15 & , & 32 & , & 67 & , & 138 & , & 281 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 \\
 + 1 & & + 2 & & + 3 & & + 4 & & + 5 & &
 \end{array}$$

$2(x) + 1 = 15$
 $2x = 14$
 $x = \frac{14}{2}$
 $x = 7$

Q3. 6, 9, 13, 19, 29, ?

(A) 52

(B) 43

(C) 42

(D) 45

(E) 47

$$\begin{array}{ccccccccc}
 6 & , & 9 & , & 13 & , & 19 & , & 29 & , & (?) \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 \\
 - 3 & & - 5 & & - 7 & & - 9 & & - 11 & &
 \end{array}
 \rightarrow (29 \times 2) - 11 = 58 - 11 = 47$$

OR

$$\begin{array}{ccccccccc}
 6 & , & 9 & , & 13 & , & 19 & , & 29 & , & (?) \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 + 3 & & + 4 & & + 6 & & + 10 & & + 18 & &
 \end{array}
 \rightarrow 29 + 18 = 47$$

$\underbrace{\quad\quad}_1 \quad \underbrace{\quad\quad}_2 \quad \underbrace{\quad\quad}_4 \quad \underbrace{\quad\quad}_8$

{ Double Difference }

Pattern 22 :- $\div a \pm b$ Rule :- Divide then add / subtract a fixed number.Examples,

$$\begin{array}{ccccccccc}
 100 & , & 52 & , & 28 & , & 16 & , & 10 & , & 7 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 \\
 + 2 & & + 2 & & + 2 & & + 2 & & + 2 & & + 2
 \end{array}$$

$$\begin{array}{ccccccccc}
 80 & , & 39 & , & 18.5 & , & 8.25 & , & 3.125 & , & 0.5625 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 \\
 - 1 & & - 1 & & - 1 & & - 1 & & - 1 & & - 1
 \end{array}$$

Practice Questions:

Q1. 70, 38, 22, 14, 10, ?

(A) 6

(B) 7

☒ (C) 8

(D) 9

(E) 10

$$\begin{array}{ccccccccc}
 70 & , & 38 & , & 22 & , & 14 & , & 10 & , & ? \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 \\
 + 3 & & + 3 & & + 3 & & + 3 & & + 3 & & + 3
 \end{array}$$

$(10 \div 2) + 3 = 5 + 3 = 8$

Q3. ?, 140, 68, 32, 14, 5

(A) 272

(B) 280

☒ (C) 284

(D) 288

(E) 292

$$\begin{array}{ccccccccc}
 ? & , & 140 & , & 68 & , & 32 & , & 14 & , & 5 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 \\
 - 2 & & - 2 & & - 2 & & - 2 & & - 2 & & - 2
 \end{array}$$

$(\frac{x}{2}) - 2 = 140 \Rightarrow \frac{x}{2} = 142 \Rightarrow x = 142 \times 2 = 284$

Q3. 246, 84, 30, 12, 6, ?

(A) 2

(B) 3

☒ (C) 4

(D) 5

(E) 6

$$\begin{array}{ccccccccc}
 246 & , & 84 & , & 30 & , & 12 & , & 6 & , & ? \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \div 3 & & \div 3 & & \div 3 & & \div 3 & & \div 3 & & \div 3 \\
 + 2 & & + 2 & & + 2 & & + 2 & & + 2 & & + 2
 \end{array}$$

$(6 \div 3) + 2 = 2 + 2 = 4$

Q4. 13285, 2660, 535, ?, 25, 8

(A) 105

(B) 115

☒ (C) 110

(D) 120

(E) 125

$$\begin{array}{ccccccccc}
 13285 & , & 2660 & , & 535 & , & ? & , & 25 & , & 8 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \div 5 & & \div 5 & & \div 5 & & \div 5 & & \div 5 & & \div 5 \\
 + 3 & & + 3 & & + 3 & & + 3 & & + 3 & & + 3
 \end{array}$$

$(535 \div 5) + 3 = 107 + 3 = 110$

Pattern 23 :- Alternating $\times$ and $\div$

Rule :- Multiply and divide alternately
{often by consecutive numbers}

Examples,

$$15, 30, 10, 40, 8, 46$$

$\times 2 \quad \div 3 \quad \times 4 \quad \div 5 \quad \times 6$

$$48, 24, 72, 18, 90, 15$$

$\div 2 \quad \times 3 \quad \div 4 \quad \times 5 \quad \div 6$

Q1. 150, 300, 100, 400, 80, ?

(A) 482

(B) 492

☒ (C) 480

(D) 478

(E) 487

$$150, 300, 100, 400, 80, ?$$

$\times 2 \quad \div 3 \quad \times 4 \quad \div 5 \quad \times 6$

$80 \times 6 = 480$

Q2. 224, 112, 336, 84, ?, 70

(A) 417

(B) 380

(C) 422

☒ (D) 420

(E) 413

$$224, 112, 336, 84, ?, 70$$

$\div 2 \quad \times 3 \quad \div 4 \quad \times 5 \quad \div 6$

$84 \times 5 = 420$

Q3. 48, 240, 40, 280, ?, 315

(A) 45

☒ (B) 35

(C) 75

(D) 56

(E) 105

$$48, 240, 40, 280, ?, 315$$

$\times 5 \quad \div 6 \quad \times 7 \quad \div 8 \quad \times 9$

$280 \div 8 = 35$

Pattern 24 → Alternate Operations

{ Repeat a pair of operations }

Rule :- Two operations repeat in cycle

Example,

• 4 , 9 , 18 , 23 , 46 , 51

$$\begin{array}{ccccccccc} & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} \\ & +5 & & \times 2 & & +5 & & \times 2 & & +5 \end{array}$$

• 30 , 27 , 54 , 51 , 102 , 99

$$\begin{array}{ccccccccc} & \text{↘} & & \text{↘} & & \text{↘} & & \text{↘} & & \text{↘} \\ & -3 & & \times 2 & & -3 & & \times 2 & & -3 \end{array}$$

{ Alternate (-3) and ($\times 2$) }

Practice Questions:

Q1. 18, 25, 75, 82, 246, ?

(A) 268

(B) 260

(C) 250

(D) 253

(E) 241

18 , 25 , 75 , 82 , 246 , ?

$$\begin{array}{ccccccccc} & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} \\ & +7 & & \times 3 & & +7 & & \times 3 & & +7 \end{array}$$

$246 + 7 = 253$

Q2. 14, 22, 44, 52, ?, 112

(A) 102

(B) 114

(C) 103

(D) 104

(E) 89

14 , 22 , 44 , 52 , ? , 112

$$\begin{array}{ccccccccc} & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} & & \text{↗} \\ & +8 & & \times 2 & & +8 & & \times 2 & & +8 \end{array}$$

$52 \times 2 = 104$

Q3. 20, 18, 54, 52, ?, 154

(A) 152

(B) 156

(C) 158

(D) 154

(E) 160

20 , 18 , 54 , 52 , ? , 154

$$\begin{array}{ccccccccc} & \text{↘} & & \text{↘} & & \text{↘} & & \text{↘} & & \text{↘} \\ & -2 & & \times 3 & & -2 & & \times 3 & & -2 \end{array}$$

$52 \times 3 = 156$

Q4. 8, 5, 25, 22, ?, 107

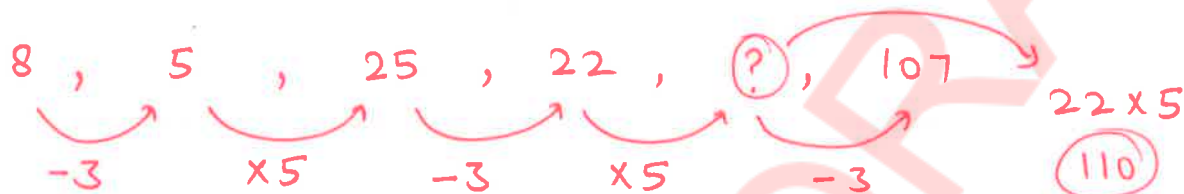
(A) 110

(B) 105

(C) 120

(D) 115

(E) 125



Pattern 25 :- Digit Sum Addition

Rule $\rightarrow$ Next term = term + sum of digits

Sum of Digits like, $\rightarrow$

$$13 = 1 + 3 = 4$$

$$17 = 1 + 7 = 8$$

$$32 = 3 + 2 = 5$$

$$25 = 2 + 5 = 7$$

Example, (1+3) (1+7) (2+5) (3+2) (3+7) (4+7)

• 13, 17, 25, 32, 37, 47

Arrows indicate the pattern: $+4$, $+8$, $+7$, $+7$, $+10$

(2+1) (2+4) (3+0) (3+3) (3+9) (5+1)

• 21, 24, 30, 33, 39, 51

Arrows indicate the pattern: $+3$, $+6$, $+3$, $+6$, $+12$

(2+1) (2+4) (3+0) (3+3) (3+9)

{Sum of Digits}

Practice Questions:

Q1. 48, 60, 66, 78, 93, ?

(A) 105

(B) 93

(C) 102

(D) 115

(E) 110

48, 60, 66, 78, 93, ? $\rightarrow 93 + 12 = 105$

$+12$ $+6$ $+12$ $+15$ $+12$
 $(4+8)$ $(6+0)$ $(6+6)$ $(7+8)$ $(9+3)$

Q2. 49, 62, 70, 77, ?, 101

(A) 92

(B) 76

(C) 101

(D) 91

(E) 81

49, 62, 70, 77, ?, 101 $\rightarrow 77 + 14 = 91$

$+13$ $+8$ $+7$ $+14$ $+10$
 $(4+9)$ $(6+2)$ $(7+0)$ $(7+7)$ $(9+1)$

Q3. ?, 34, 41, 46, 56, 67

(A) 25

(B) 26

(C) 27

(D) 30

(E) 22

26 $\leftarrow$?, 34, 41, 46, 56, 67

$+8$ $+7$ $+5$ $+10$ $+11$
 $(2+6)$ $(3+4)$ $(4+1)$ $(4+6)$ $(5+6)$

Q4. 10, 11, 13, 17, ?, 32

(A) 26

(B) 29

(C) 21

(D) 30

(E) 25

10, 11, 13, 17, ?, 32 $\rightarrow 17 + 8 = 25$

$+1$ $+2$ $+4$ $+8$ $+7$
 $(1+0)$ $(1+1)$ $(1+3)$ $(1+7)$ $(2+5)$

PRACTICE QUESTIONS (ALL PATTERNS)

Q1. 2, 4, 12, 48, 240, ?

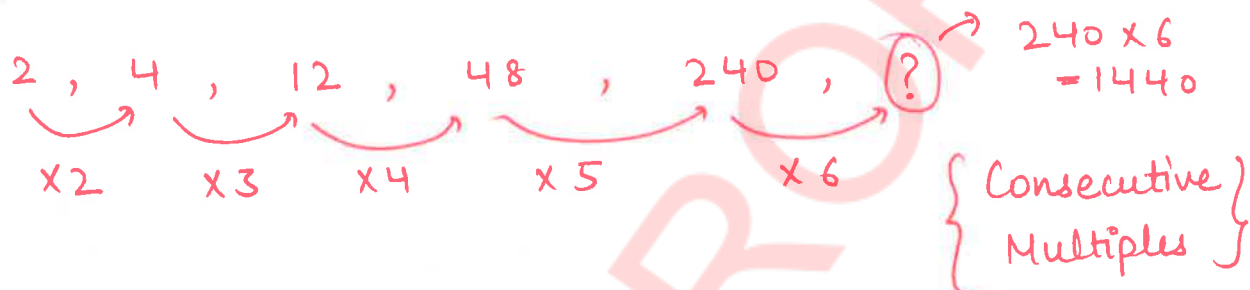
(A) 1320

(B) 1360

(C) 1400

(D) 1580

(E) 1440



Q2. 4, 27, 16, 125, 36, 343, ?

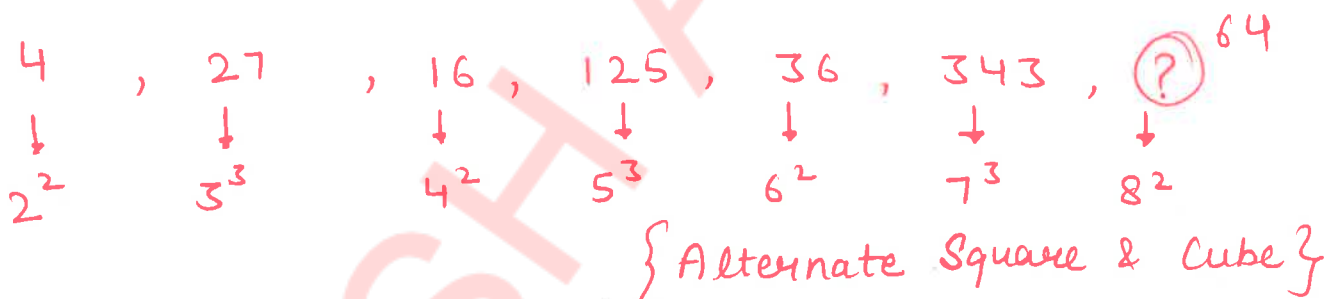
(A) 49

(B) 81

(C) 64

(D) 72

(E) None of these



Q3. 2, 6, 12, 20, 30, ?, 56

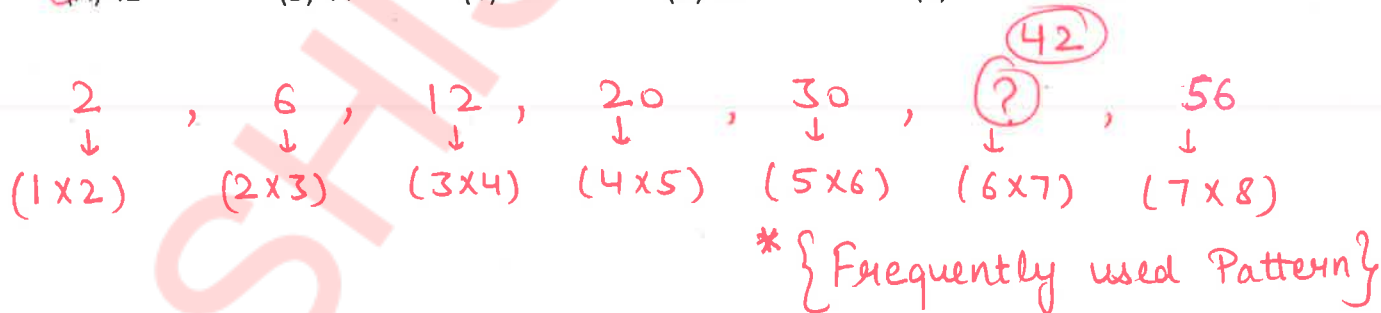
(A) 42

(B) 44

(C) 46

(D) 48

(E) None of these



Q4. 48, 24, 72, 18, 90, 15, ?

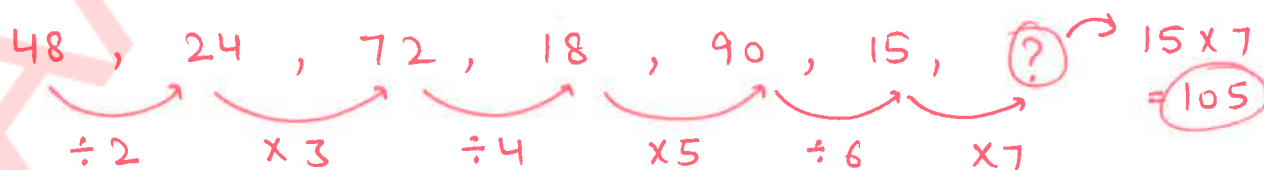
(A) 96

(B) 105

(C) 108

(D) 120

(E) None of these



Alternate Division & Multiplication

{ Fibonacci Series }

Q5. 6, 9, 15, 24, 39, 63, ?

(A) 96

(B) 99

(C) 100

☒ (D) 102

(E) None of these

$$\begin{array}{ccccccccc}
 6 & , & 9 & , & 15 & , & 24 & , & 39 & , & 63 & , & ? & \rightarrow 102 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 & & & & (6+9) & & (9+15) & & (15+24) & & (24+39) & & (39+63)
 \end{array}$$

Q6. ?, 14, 28, 56, 112, 224

(A) 4

(B) 5

(C) 6

☒ (D) 7

(E) None of these

$$\begin{array}{ccccccccc}
 7 & \leftarrow ? & , & 14 & , & 28 & , & 56 & , & 112 & , & 224 \\
 & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\
 & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2
 \end{array}$$

{ Fixed Number Multiplier }

Q7. 1, 2, 10, 37, 101, ?, 442

(A) 210

☒ (B) 226

(C) 224

(D) 228

(E) None of these

$$\begin{array}{ccccccccc}
 1 & , & 2 & , & 10 & , & 37 & , & 101 & , & ? & , & 442 \\
 \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\
 +1 & & +8 & & +27 & & +64 & & +125 & & +216 \\
 (1^3) & & (2^3) & & (3^3) & & (4^3) & & (5^3) & & (6^3)
 \end{array}$$

$$101 + 125 = 226$$

Q8. 2, 5, 16, 65, 326, ?

☒ (A) 1957

(B) 1857

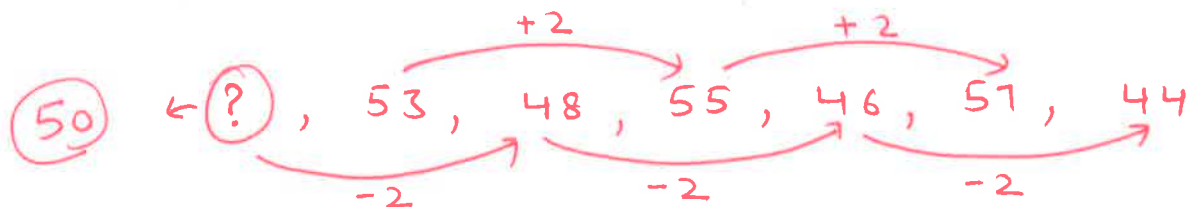
(C) 2057

(D) 1975

(E) None of these

$$\begin{array}{ccccccccc}
 2 & , & 5 & , & 16 & , & 65 & , & 326 & , & ? & \rightarrow (326 \times 6) + 1 \\
 \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\
 \times 2 & & \times 3 & & \times 4 & & \times 5 & & \times 6 \\
 +1 & & +1 & & +1 & & +1 & & +1
 \end{array}$$

$$1956 + 1 = 1957$$



Q9. ?, 53, 48, 55, 46, 57, 44

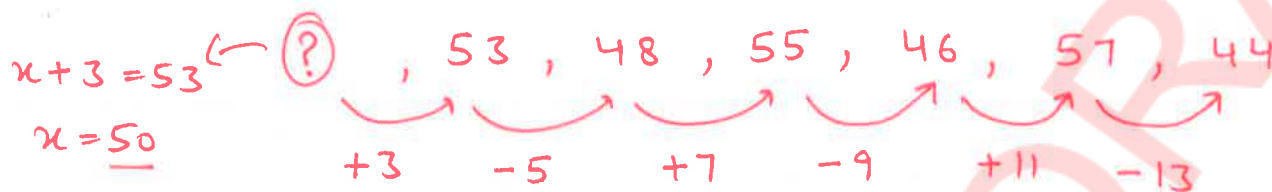
(A) 48

(B) 46

(C) 50

(D) 52

(E) None of these



Q10. 2, 3, 7, 13, 27, 53, ?

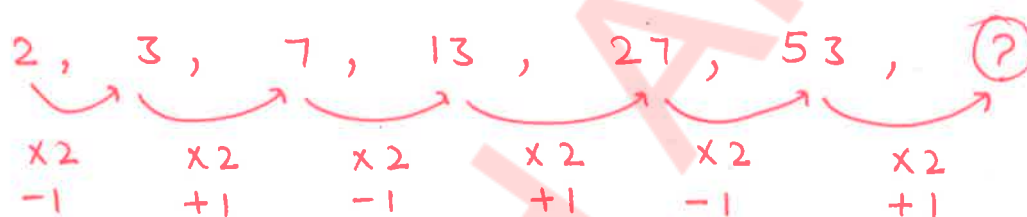
(A) 103

(B) 105

(C) 109

(D) 111

(E) None of these



Q11. ?, 4, 8, 17, 33, 58, 94

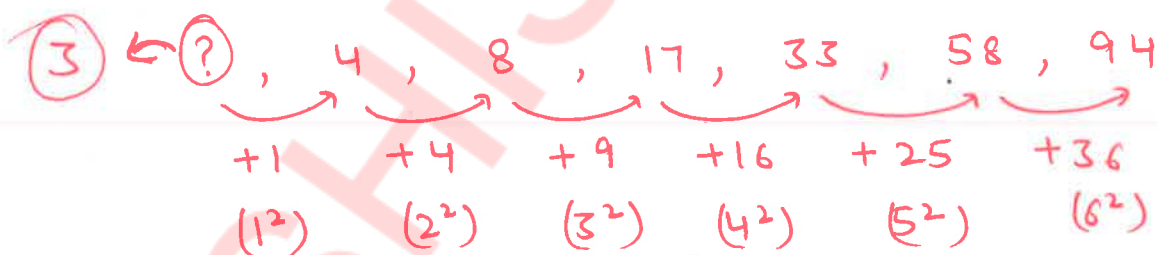
(A) 3

(B) 5

(C) 2

(D) 4

(E) None of these



Q12. 10, 13, 19, 31, 55, ?

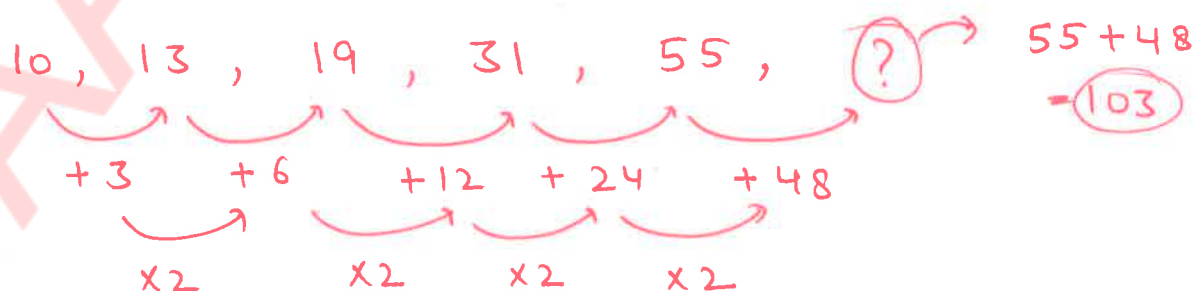
(A) 99

(B) 101

(C) 105

(D) 103

(E) None of these



Q13. 3, 8, 19, 42, 89, ?

(A) 176

☒ (B) 184

(C) 188

(D) 192

(E) None of these

$$\begin{array}{ccccccccc}
 3 & , & 8 & , & 19 & , & 42 & , & 89 & , & ? \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 & & \times 2 \\
 + 2 & & + 3 & & + 4 & & + 5 & & + 6 & & + 6
 \end{array}$$

$(89 \times 2) + 6$
 $178 + 6$
 184

Q14. 80, 40, 60, 30, 45, ?, 33.75

(A) 20.5

(B) 21.5

(C) 23.5

☒ (D) 22.5

(E) None of these

$$\begin{array}{ccccccccc}
 80 & , & 40 & , & 60 & , & 30 & , & 45 & , & ? & , & 33.75 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \div 2 & & \times 1.5 & & \div 2 & & \times 1.5 & & \div 2 & & \times 1.5 & & \times 1.5
 \end{array}$$

$$\{45 \div 2 = 22.5\}$$

Q15. 10, 11, 15, 42, 58, ?

(A) 178

☒ (B) 183

(C) 189

(D) 193

(E) None of these

$$\begin{array}{ccccccccc}
 10 & , & 11 & , & 15 & , & 42 & , & 58 & , & ? \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 + 1 & & + 4 & & + 27 & & + 16 & & + 125 & & + 125 \\
 (1^3) & & (2^2) & & (3^3) & & (4^2) & & (5^3) & & (6^3)
 \end{array}$$

$58 + 125$
 183

Q16. ?, 32, 32, 48, 96, 240

(A) 16

(B) 32

☒ (C) 64

(D) 96

(E) None of these

$$\begin{array}{ccccccccc}
 ? & , & 32 & , & 32 & , & 48 & , & 96 & , & 240 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\
 \times 0.5 & & \times 1 & & \times 1.5 & & \times 2 & & \times 2.5 & & \times 3
 \end{array}$$

$$x \times 0.5 = 32$$

$$x \times \frac{1}{2} = 32$$

$$x = 64$$

Q17. 6250, 625, 125, 50, ?, 64

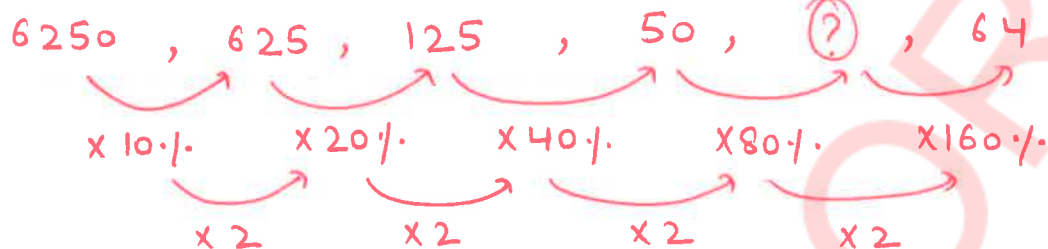
(A) 32

(B) 36

(C) 48

(D) 40

(E) None of these



Q18. 5, 6, 9, 15, 25, 40, ?

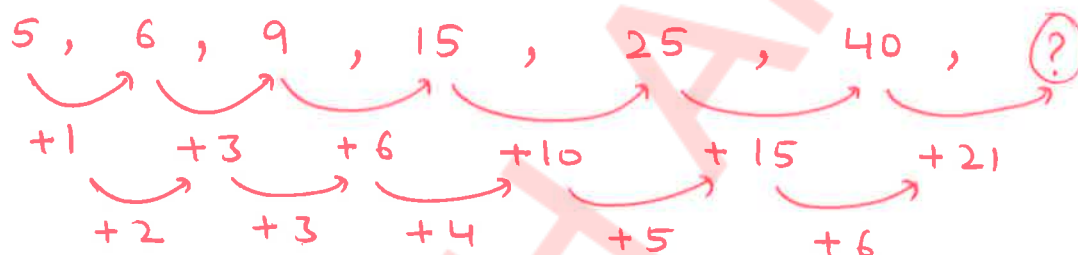
(A) 60

(B) 62

(C) 61

(D) 63

(E) None of these



Q19. 5, 7, 21, 25, 125, 131, ?

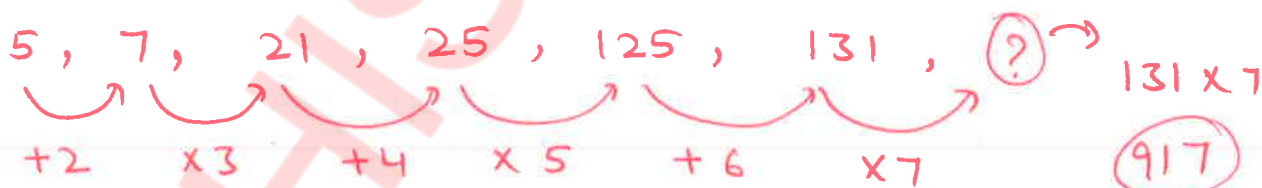
(A) 901

(B) 917

(C) 921

(D) 933

(E) None of these



Q20. 10, 29, 85, 252, ?, 2251

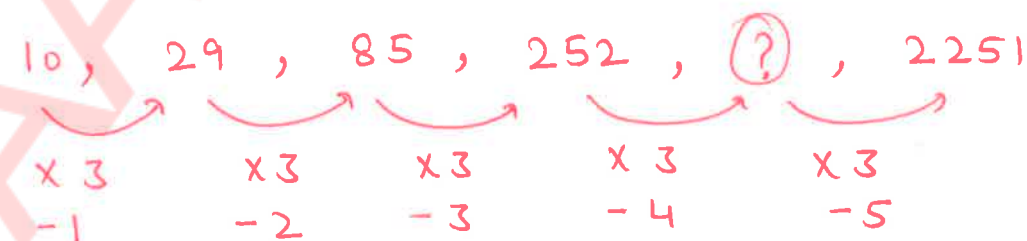
(A) 748

(B) 752

(C) 754

(D) 756

(E) None of these



$$(252 \times 3) - 4$$

$$756 - 4 = 752$$

Q1: 13, 32, 59, 94, 137, ?

(A) 186

(B) 188

(C) 189

(D) 190

(E) 192

$$\begin{array}{ccccccccc}
 13 & , & 32 & , & 59 & , & 94 & , & 137 & , & (?) & \rightarrow & 137 + 51 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \textcircled{188} \\
 +19 & & +27 & & +35 & & +43 & & +51 & & & & \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \\
 +8 & & +8 & & +8 & & +8 & & & & & &
 \end{array}$$

Q2: 513, 257, 129, 65, 33, ?

(A) 15

(B) 16

(C) 17

(D) 18

(E) 19

$$\begin{array}{ccccccccc}
 513 & , & 257 & , & 129 & , & 65 & , & 33 & , & (?) & \rightarrow & \left(\frac{33}{2}\right) + 0.5 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \\
 \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \div 2 & & \\
 +0.5 & & +0.5 & & +0.5 & & +0.5 & & +0.5 & & +0.5 & & \\
 & & & & & & & & & & & & 16.5 + 0.5 \\
 & & & & & & & & & & & & = \textcircled{17}
 \end{array}$$

Q3: 62, 123, 214, 341, 510, ?

(A) 724

(B) 729

(C) 728

(D) 727

(E) 731

$$\begin{array}{ccccccccc}
 62 & , & 123 & , & 214 & , & 341 & , & 510 & , & (?) & \rightarrow & \underline{\underline{727}} \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
 64-2 & & 125-2 & & 216-2 & & 343-2 & & 512-2 & & 729-2 & & \\
 (4^3-2) & & (5^3-2) & & (6^3-2) & & (7^3-2) & & (8^3-2) & & (9^3-2) & &
 \end{array}$$

Q4: 32, 50, 72, 98, ?, 162

(A) 128

(B) 112

(C) 144

(D) 150

(E) None of these

$$\begin{array}{ccccccccc}
 32 & , & 50 & , & 72 & , & 98 & , & (?) & , & 162 & & 98 + 30 \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \textcircled{128} \\
 +18 & & +22 & & +26 & & +30 & & +34 & & & & \\
 \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \swarrow & & \\
 +4 & & +4 & & +4 & & +4 & & & & & &
 \end{array}$$